(1)

x_i :	-1	0	1	2
y_i :	2	2	2	8

Og45

Interpolationspolynom

a) Lagrange

b) Newton

a)
$$L_j(x) = \prod_{i \neq j} \frac{x - x_i}{x_j - x_i}$$

$$p(x) = \sum y_j L_j(x)$$

x_i : -1, 0, 1, 2

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \frac{x - x_2}{x_0 - x_2} \frac{x - x_3}{x_0 \quad x_3}$$

$$= \frac{x - 0}{-1 - 0} \frac{x - 1}{-1 - 1} \frac{x - 2}{-1 - 2}$$

$$L_1(x) = \frac{x - x_0}{x_1 - x_0} \frac{x - x_2}{x_1 - x_2} \frac{x - x_3}{x_1 - x_3}$$

$$= \frac{x - (-1)}{0 - (-1)} \frac{x - 1}{0 - 1} \frac{x - 2}{0 - 2}$$

$$L_2(x) = \frac{x - (-1)}{1 - (-1)} \frac{x - 0}{1 - 0} \frac{x - 2}{1 - 2}$$

$$L_3(x) = \frac{x - (-1)}{2 - (-1)} \frac{x - 0}{2 - 0} \frac{x - 1}{2 - 1}$$

$$p(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x)$$

$$= \ldots = 2 - x + x^3$$

b) <u>Newton</u>

x_i	y_i			
-1	2			
0	2	$\frac{2-2}{0-(-1)} = 0$		
1	2	$\frac{2-2}{1-0} = 0$	$\frac{0-0}{1-(-1)} = 0$	
2	8	$\frac{8-2}{2-1} = 6$	$\frac{6-0}{2-0} = 3$	$\underbrace{\frac{3-0}{2-(-1)}}_{= 1}$

$$p(x) = 2 + 0\,(x-x_0) + 0\,(x-x_0)(x-x_1) + 1\,(x-x_0)(x-x_1)(x-x_2)$$

$$= 2 + \cancel{0\,(x-(-1))} + \cancel{0\,(x-(-1))}(x-0) + 1\,(x-(-1))(x-0)(x-1)$$

$$= 2 + (x+1)\,x\,(x-1)$$

$$= 2 + (x^2-1)\,x$$

$$= 2 + x^3 - x$$